Banking Analytics Data Warehouse & Power BI Dashboard

Project Overview

Design and implement an end-to-end Banking Analytics Data Warehouse that transforms raw banking data into business-ready analytical datasets and delivers insights through interactive Power BI reporting.

Phase 1 – Data Ingestion (Landing Layer)

Objective

Load raw banking datasets into the Landing Layer without modification.

Deliverables

- transaction_raw
- clients_raw
- cards_raw
- mcc_codes_raw
- fraud_labels_raw

Activities

- Create Landing schema
- Import CSV and JSON files
- Validate data loads

Phase 2 – Data Transformation (Core Layer)

Objective

Clean and standardize source data for downstream analytics.

Deliverables

- transaction
- clients
- cards
- mcc_code
- fraud_labels

Activities

- Data cleansing
- Type conversions
- Standardization
- Business rule implementation

Phase 3 – Analytics Data Modeling (Mart Layer)

Objective

Create reporting-ready datasets for business analysis.

Deliverables

- fact_transactions
- customer_summary
- customer_segments
- merchant_summary
- card_performance
- state_summary
- category_summary
- monthly_transaction_trend
- fraud_summary
- dim_customer
- dim_card
- dim_mcc

Activities

- Create analytical views
- Build aggregations
- Optimize reporting queries

Phase 4 – End-to-End Data Warehouse Architecture

Objective

Document the complete flow of data from source systems to reporting.

Deliverables

- End-to-End Data Warehouse Architecture Diagram
- Data Flow Diagram
- Data Lineage Documentation

Activities

- Document source-to-report flow
- Document layer dependencies
- Document reporting dependencies

Phase 5 – Power BI Dashboard Development

Objective

Create an executive-level banking analytics dashboard.

KPIs

- Total Transactions
- Total Transaction Amount
- Average Transaction Amount
- Total Customers
- Fraud Rate %

Visuals

- Transaction Trend
- Transactions by Location
- Customer Segments by Income Band
- Card Usage Distribution
- Merchant Category Analysis

Filters

- Year

- Month
- Location
- Card Brand

Phase 6 – Testing and Validation

Objective

Ensure reporting accuracy.

Activities

- Validate data transformations
- Reconcile metrics
- Verify Power BI calculations

Phase 7 – GitHub Portfolio Publication

Objective

Publish the project as a professional portfolio.

Deliverables

- SQL Scripts
- Architecture Diagram
- Power BI Dashboard
- Documentation
- README

Technologies Used

- SQL Server
- T-SQL
- Data Warehouse Architecture
- Power BI
- Draw.io

- GitHub